SG No. 216 T_{d}^2 $F\bar{4}3m$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$
* Wyckoff site: `4a`, site symmetry: `-43m`

Table 1: Wyckoff bond: `12a@4a`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X,\,0,\,0]$	[0, 0, 0]	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0,\,X,\,0]$	[0, 0, 0]	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0,\,0,\,X]$	[0, 0, 0]	`[9,-10,-11,12,21,-22,-23,24]`

Table 2: Wyckoff bond: `16b@4a`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	[0, 0, 0]	`[1,5,9,13,17,21]`
2	$[-X,\,-X,\,X]$	[0, 0, 0]	`[2,7,12,14,19,24]`
3	$[-X,\,X,\,-X]$	[0, 0, 0]	`[3,8,10,16,18,23]`
4	$[X,\,-X,\,-X]$	[0, 0, 0]	`[4,6,11,15,20,22]`

Table 3: Wyckoff bond: `24c@4a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X,\,X,\,0]$	[0, 0, 0]	`[1,-2,13,-14]`
2	$[-X,\,X,\,0]$	[0, 0, 0]	`[3,-4,-15,16]`
3	$[0,\,X,\,X]$	[0, 0, 0]	`[5,-6,21,-22]`
4	$[0,\,-X,\,X]$	[0, 0, 0]	`[7,-8,-23,24]`
5	$[X,\,0,\,X]$	[0, 0, 0]	`[9,-10,17,-18]`
6	$[X,\,0,\,-X]$	[0, 0, 0]	`[11,-12,-19,20]`

Table 4: Wyckoff bond: `48d@4a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X,\,X,\,Y]$	[0, 0, 0]	`[1,13]`
2	$[-X,\,-X,\,Y]$	[0, 0, 0]	`[2,14]`
3	$[-X,\,X,\,-Y]$	[0, 0, 0]	`[3,16]`
4	$[X,\,-X,\,-Y]$	[0, 0, 0]	`[4,15]`
5	$[Y,\,X,\,X]$	[0, 0, 0]	`[5,21]`
6	$[Y,\,-X,\,-X]$	[0, 0, 0]	`[6,22]`
7	$[-Y,\,-X,\,X]$	[0, 0, 0]	`[7,24]`
8	$[-Y,\,X,\,-X]$	[0, 0, 0]	`[8,23]`
9	$[X,\,Y,\,X]$	[0, 0, 0]	`[9,17]`
10	$[-X,\,Y,\,-X]$	[0, 0, 0]	`[10,18]`
11	$[X,\,-Y,\,-X]$	[0, 0, 0]	`[11,20]`
12	$[-X,\,-Y,\,X]$	[0, 0, 0]	`[12,19]`

Table 5: Wyckoff bond: `48e@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	[0, 0, 0]	`[1,-2]`
2	$[-X, Y, 0]$	[0, 0, 0]	`[3,-4]`
3	$[0, X, Y]$	[0, 0, 0]	`[5,-6]`
4	$[0, -X, Y]$	[0, 0, 0]	`[7,-8]`
5	$[Y, 0, X]$	[0, 0, 0]	`[9,-10]`
6	$[Y, 0, -X]$	[0, 0, 0]	`[11,-12]`
7	$[Y, X, 0]$	[0, 0, 0]	`[13,-14]`
8	$[Y, -X, 0]$	[0, 0, 0]	`[15,-16]`
9	$[X, 0, Y]$	[0, 0, 0]	`[17,-18]`
10	$[-X, 0, Y]$	[0, 0, 0]	`[19,-20]`
11	$[0, Y, X]$	[0, 0, 0]	`[21,-22]`
12	$[0, Y, -X]$	[0, 0, 0]	`[23,-24]`

Table 6: Wyckoff bond: `96f@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	[0, 0, 0]	`[1]`
2	$[-X, -Y, Z]$	[0, 0, 0]	`[2]`
3	$[-X, Y, -Z]$	[0, 0, 0]	`[3]`
4	$[X, -Y, -Z]$	[0, 0, 0]	`[4]`
5	$[Z, X, Y]$	[0, 0, 0]	`[5]`
6	$[Z, -X, -Y]$	[0, 0, 0]	`[6]`
7	$[-Z, -X, Y]$	[0, 0, 0]	`[7]`
8	$[-Z, X, -Y]$	[0, 0, 0]	`[8]`
9	$[Y, Z, X]$	[0, 0, 0]	`[9]`
10	$[-Y, Z, -X]$	[0, 0, 0]	`[10]`
11	$[Y, -Z, -X]$	[0, 0, 0]	`[11]`
12	$[-Y, -Z, X]$	[0, 0, 0]	`[12]`
13	$[Y, X, Z]$	[0, 0, 0]	`[13]`
14	$[-Y, -X, Z]$	[0, 0, 0]	`[14]`
15	$[Y, -X, -Z]$	[0, 0, 0]	`[15]`
16	$[-Y, X, -Z]$	[0, 0, 0]	`[16]`
17	$[X, Z, Y]$	[0, 0, 0]	`[17]`
18	$[-X, Z, -Y]$	[0, 0, 0]	`[18]`
19	$[-X, -Z, Y]$	[0, 0, 0]	`[19]`
20	$[X, -Z, -Y]$	[0, 0, 0]	`[20]`
21	$[Z, Y, X]$	[0, 0, 0]	`[21]`
22	$[Z, -Y, -X]$	[0, 0, 0]	`[22]`
23	$[-Z, Y, -X]$	[0, 0, 0]	`[23]`
24	$[-Z, -Y, X]$	[0, 0, 0]	`[24]`

* Wyckoff site: `4b`, site symmetry: `-43m`

Table 7: Wyckoff bond: `12a@4b`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-10,-11,12,21,-22,-23,24]`

Table 8: Wyckoff bond: `16b@4b`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,5,9,13,17,21]`
2	$[-X, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2,7,12,14,19,24]`
3	$[-X, X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,8,10,16,18,23]`
4	$[X, -X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4,6,11,15,20,22]`

Table 9: Wyckoff bond: `24c@4b`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,13,-14]`
2	$[-X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-4,-15,16]`
3	$[0, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-6,21,-22]`
4	$[0, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[7,-8,-23,24]`
5	$[X, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-10,17,-18]`
6	$[X, 0, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[11,-12,-19,20]`

Table 10: Wyckoff bond: `48d@4b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,13]`
2	$[-X, -X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2,14]`
3	$[-X, X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,16]`
4	$[X, -X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4,15]`
5	$[Y, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,21]`
6	$[Y, -X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[6,22]`
7	$[-Y, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[7,24]`
8	$[-Y, X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[8,23]`
9	$[X, Y, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,17]`
10	$[-X, Y, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[10,18]`
11	$[X, -Y, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[11,20]`
12	$[-X, -Y, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[12,19]`

Table 11: Wyckoff bond: `48e@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2]`

continued ...

Table 11

No.	vector	center	mapping
2	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,-4]
3	$[0, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[5,-6]
4	$[0, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[7,-8]
5	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,-10]
6	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[11,-12]
7	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[13,-14]
8	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[15,-16]
9	$[X, 0, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,-18]
10	$[-X, 0, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[19,-20]
11	$[0, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[21,-22]
12	$[0, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[23,-24]

Table 12: Wyckoff bond: 96f@4b, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[12]
13	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[13]
14	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[14]
15	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[15]
16	$[-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[16]
17	$[X, Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17]
18	$[-X, Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[19]
20	$[X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[20]
21	$[Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[21]
22	$[Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[22]
23	$[-Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[23]
24	$[-Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[24]

* Wyckoff site: 4c, site symmetry: -43m

Table 13: Wyckoff bond: `12a@4c`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[9,-10,-11,12,21,-22,-23,24]`

Table 14: Wyckoff bond: `16b@4c`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,5,9,13,17,21]`
2	$[-X, -X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,7,12,14,19,24]`
3	$[-X, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3,8,10,16,18,23]`
4	$[X, -X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,6,11,15,20,22]`

Table 15: Wyckoff bond: `24c@4c`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2,13,-14]`
2	$[-X, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3,-4,-15,16]`
3	$[0, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,-6,21,-22]`
4	$[0, -X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[7,-8,-23,24]`
5	$[X, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[9,-10,17,-18]`
6	$[X, 0, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[11,-12,-19,20]`

Table 16: Wyckoff bond: `48d@4c`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,13]`
2	$[-X, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,14]`
3	$[-X, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3,16]`
4	$[X, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,15]`
5	$[Y, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,21]`
6	$[Y, -X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[6,22]`
7	$[-Y, -X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[7,24]`
8	$[-Y, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[8,23]`
9	$[X, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[9,17]`
10	$[-X, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[10,18]`
11	$[X, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[11,20]`
12	$[-X, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[12,19]`

Table 17: Wyckoff bond: `48e@4c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2]`

continued ...

Table 17

No.	vector	center	mapping
2	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[3,-4]
3	$[0, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,-6]
4	$[0, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[7,-8]
5	$[Y, 0, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[9,-10]
6	$[Y, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[11,-12]
7	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[13,-14]
8	$[Y, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[15,-16]
9	$[X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[17,-18]
10	$[-X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[19,-20]
11	$[0, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[21,-22]
12	$[0, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[23,-24]

Table 18: Wyckoff bond: 96f@4c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[12]
13	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[13]
14	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[14]
15	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[15]
16	$[-Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[16]
17	$[X, Z, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[17]
18	$[-X, Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[18]
19	$[-X, -Z, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[19]
20	$[X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[20]
21	$[Z, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[21]
22	$[Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[22]
23	$[-Z, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[23]
24	$[-Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[24]

* Wyckoff site: 4d, site symmetry: -43m

Table 19: Wyckoff bond: `12a@4d`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[9,-10,-11,12,21,-22,-23,24]`

Table 20: Wyckoff bond: `16b@4d`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1,5,9,13,17,21]`
2	$[-X, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[2,7,12,14,19,24]`
3	$[-X, X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,8,10,16,18,23]`
4	$[X, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,6,11,15,20,22]`

Table 21: Wyckoff bond: `24c@4d`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1,-2,13,-14]`
2	$[-X, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-4,-15,16]`
3	$[0, X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[5,-6,21,-22]`
4	$[0, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[7,-8,-23,24]`
5	$[X, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[9,-10,17,-18]`
6	$[X, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[11,-12,-19,20]`

Table 22: Wyckoff bond: `48d@4d`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1,13]`
2	$[-X, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[2,14]`
3	$[-X, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,16]`
4	$[X, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,15]`
5	$[Y, X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[5,21]`
6	$[Y, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[6,22]`
7	$[-Y, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[7,24]`
8	$[-Y, X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[8,23]`
9	$[X, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[9,17]`
10	$[-X, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[10,18]`
11	$[X, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[11,20]`
12	$[-X, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[12,19]`

Table 23: Wyckoff bond: `48e@4d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[1,-2]`

continued ...

Table 23

No.	vector	center	mapping
2	$[-X, Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[3,-4]
3	$[0, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[5,-6]
4	$[0, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[7,-8]
5	$[Y, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[9,-10]
6	$[Y, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[11,-12]
7	$[Y, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[13,-14]
8	$[Y, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[15,-16]
9	$[X, 0, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[17,-18]
10	$[-X, 0, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[19,-20]
11	$[0, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[21,-22]
12	$[0, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[23,-24]

Table 24: Wyckoff bond: 96f@4d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[1]
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[2]
3	$[-X, Y, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[3]
4	$[X, -Y, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[4]
5	$[Z, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[5]
6	$[Z, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[6]
7	$[-Z, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[7]
8	$[-Z, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[8]
9	$[Y, Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[9]
10	$[-Y, Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[10]
11	$[Y, -Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[11]
12	$[-Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[12]
13	$[Y, X, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[13]
14	$[-Y, -X, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[14]
15	$[Y, -X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[15]
16	$[-Y, X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[16]
17	$[X, Z, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[17]
18	$[-X, Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[18]
19	$[-X, -Z, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[19]
20	$[X, -Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[20]
21	$[Z, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[21]
22	$[Z, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[22]
23	$[-Z, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[23]
24	$[-Z, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[24]

* Wyckoff site: 16e, site symmetry: .3m

Table 25: Wyckoff bond: `16a@16e`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9,13,17,21]`
2	$[-X, -X, X]$	$[-x, -x, x]$	`[2,7,12,14,19,24]`
3	$[-X, X, -X]$	$[-x, x, -x]$	`[3,8,10,16,18,23]`
4	$[X, -X, -X]$	$[x, -x, -x]$	`[4,6,11,15,20,22]`

Table 26: Wyckoff bond: `48b@16e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, x]$	`[1,13]`
2	$[-X, -X, Y]$	$[-x, -x, x]$	`[2,14]`
3	$[-X, X, -Y]$	$[-x, x, -x]$	`[3,16]`
4	$[X, -X, -Y]$	$[x, -x, -x]$	`[4,15]`
5	$[Y, X, X]$	$[x, x, x]$	`[5,21]`
6	$[Y, -X, -X]$	$[x, -x, -x]$	`[6,22]`
7	$[-Y, -X, X]$	$[-x, -x, x]$	`[7,24]`
8	$[-Y, X, -X]$	$[-x, x, -x]$	`[8,23]`
9	$[X, Y, X]$	$[x, x, x]$	`[9,17]`
10	$[-X, Y, -X]$	$[-x, x, -x]$	`[10,18]`
11	$[X, -Y, -X]$	$[x, -x, -x]$	`[11,20]`
12	$[-X, -Y, X]$	$[-x, -x, x]$	`[12,19]`

Table 27: Wyckoff bond: `48c@16e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, x]$	`[1,-13]`
2	$[-X, X, 0]$	$[-x, -x, x]$	`[2,-14]`
3	$[-X, -X, 0]$	$[-x, x, -x]$	`[3,-16]`
4	$[X, X, 0]$	$[x, -x, -x]$	`[4,-15]`
5	$[0, X, -X]$	$[x, x, x]$	`[5,-21]`
6	$[0, -X, X]$	$[x, -x, -x]$	`[6,-22]`
7	$[0, -X, -X]$	$[-x, -x, x]$	`[7,-24]`
8	$[0, X, X]$	$[-x, x, -x]$	`[8,-23]`
9	$[-X, 0, X]$	$[x, x, x]$	`[9,-17]`
10	$[X, 0, -X]$	$[-x, x, -x]$	`[10,-18]`
11	$[-X, 0, -X]$	$[x, -x, -x]$	`[11,-20]`
12	$[X, 0, X]$	$[-x, -x, x]$	`[12,-19]`

Table 28: Wyckoff bond: `96d@16e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, x]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -x]$	`[3]`

continued ...

Table 28

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$[x, -x, -x]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$[x, -x, -x]$	[6]
7	$[-Z, -X, Y]$	$[-x, -x, x]$	[7]
8	$[-Z, X, -Y]$	$[-x, x, -x]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$[-x, x, -x]$	[10]
11	$[Y, -Z, -X]$	$[x, -x, -x]$	[11]
12	$[-Y, -Z, X]$	$[-x, -x, x]$	[12]
13	$[Y, X, Z]$	$[x, x, x]$	[13]
14	$[-Y, -X, Z]$	$[-x, -x, x]$	[14]
15	$[Y, -X, -Z]$	$[x, -x, -x]$	[15]
16	$[-Y, X, -Z]$	$[-x, x, -x]$	[16]
17	$[X, Z, Y]$	$[x, x, x]$	[17]
18	$[-X, Z, -Y]$	$[-x, x, -x]$	[18]
19	$[-X, -Z, Y]$	$[-x, -x, x]$	[19]
20	$[X, -Z, -Y]$	$[x, -x, -x]$	[20]
21	$[Z, Y, X]$	$[x, x, x]$	[21]
22	$[Z, -Y, -X]$	$[x, -x, -x]$	[22]
23	$[-Z, Y, -X]$	$[-x, x, -x]$	[23]
24	$[-Z, -Y, X]$	$[-x, -x, x]$	[24]

* Wyckoff site: 24f, site symmetry: 2.mm

Table 29: Wyckoff bond: 24a@24f, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[0, X, X]$	$[x, 0, 0]$	[1,-4,17,-20]
2	$[0, -X, X]$	$[-x, 0, 0]$	[2,-3,-18,19]
3	$[X, 0, X]$	$[0, x, 0]$	[5,-8,13,-16]
4	$[X, 0, -X]$	$[0, -x, 0]$	[6,-7,-14,15]
5	$[X, X, 0]$	$[0, 0, x]$	[9,-12,21,-24]
6	$[-X, X, 0]$	$[0, 0, -x]$	[10,-11,-22,23]

Table 30: Wyckoff bond: 24b@24f, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	[1,4,17,20]
2	$[-X, 0, 0]$	$[-x, 0, 0]$	[2,3,18,19]
3	$[0, X, 0]$	$[0, x, 0]$	[5,8,13,16]
4	$[0, -X, 0]$	$[0, -x, 0]$	[6,7,14,15]
5	$[0, 0, X]$	$[0, 0, x]$	[9,12,21,24]
6	$[0, 0, -X]$	$[0, 0, -x]$	[10,11,22,23]

Table 31: Wyckoff bond: `48c@24f`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[x, 0, 0]$	`[1,17]`
2	$[-Y, -X, X]$	$[-x, 0, 0]$	`[2,19]`
3	$[-Y, X, -X]$	$[-x, 0, 0]$	`[3,18]`
4	$[Y, -X, -X]$	$[x, 0, 0]$	`[4,20]`
5	$[X, Y, X]$	$[0, x, 0]$	`[5,13]`
6	$[X, -Y, -X]$	$[0, -x, 0]$	`[6,15]`
7	$[-X, -Y, X]$	$[0, -x, 0]$	`[7,14]`
8	$[-X, Y, -X]$	$[0, x, 0]$	`[8,16]`
9	$[X, X, Y]$	$[0, 0, x]$	`[9,21]`
10	$[-X, X, -Y]$	$[0, 0, -x]$	`[10,23]`
11	$[X, -X, -Y]$	$[0, 0, -x]$	`[11,22]`
12	$[-X, -X, Y]$	$[0, 0, x]$	`[12,24]`

Table 32: Wyckoff bond: `48d@24f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$[-x, 0, 0]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, x, 0]$	`[5,-8]`
4	$[Y, 0, -X]$	$[0, -x, 0]$	`[6,-7]`
5	$[X, Y, 0]$	$[0, 0, x]$	`[9,-12]`
6	$[-X, Y, 0]$	$[0, 0, -x]$	`[10,-11]`
7	$[X, 0, Y]$	$[0, x, 0]$	`[13,-16]`
8	$[-X, 0, Y]$	$[0, -x, 0]$	`[14,-15]`
9	$[0, Y, X]$	$[x, 0, 0]$	`[17,-20]`
10	$[0, Y, -X]$	$[-x, 0, 0]$	`[18,-19]`
11	$[Y, X, 0]$	$[0, 0, x]$	`[21,-24]`
12	$[Y, -X, 0]$	$[0, 0, -x]$	`[22,-23]`

Table 33: Wyckoff bond: `96e@24f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, x, 0]$	`[5]`
6	$[Z, -X, -Y]$	$[0, -x, 0]$	`[6]`
7	$[-Z, -X, Y]$	$[0, -x, 0]$	`[7]`
8	$[-Z, X, -Y]$	$[0, x, 0]$	`[8]`
9	$[Y, Z, X]$	$[0, 0, x]$	`[9]`
10	$[-Y, Z, -X]$	$[0, 0, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[0, 0, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[0, 0, x]$	`[12]`

continued ...

Table 33

No.	vector	center	mapping
13	$[Y, X, Z]$	$[0, x, 0]$	`[13]`
14	$[-Y, -X, Z]$	$[0, -x, 0]$	`[14]`
15	$[Y, -X, -Z]$	$[0, -x, 0]$	`[15]`
16	$[-Y, X, -Z]$	$[0, x, 0]$	`[16]`
17	$[X, Z, Y]$	$[x, 0, 0]$	`[17]`
18	$[-X, Z, -Y]$	$[-x, 0, 0]$	`[18]`
19	$[-X, -Z, Y]$	$[-x, 0, 0]$	`[19]`
20	$[X, -Z, -Y]$	$[x, 0, 0]$	`[20]`
21	$[Z, Y, X]$	$[0, 0, x]$	`[21]`
22	$[Z, -Y, -X]$	$[0, 0, -x]$	`[22]`
23	$[-Z, Y, -X]$	$[0, 0, -x]$	`[23]`
24	$[-Z, -Y, X]$	$[0, 0, x]$	`[24]`

* Wyckoff site: `24g`, site symmetry: `2.mm`

Table 34: Wyckoff bond: `24a@24g`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-4,17,-20]`
2	$[0, -X, X]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[2,-3,-18,19]`
3	$[X, 0, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,-8,13,-16]`
4	$[X, 0, -X]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	`[6,-7,-14,15]`
5	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	`[10,-11,-22,23]`

Table 35: Wyckoff bond: `24b@24g`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,17,20]`
2	$[-X, 0, 0]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,18,19]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,8,13,16]`
4	$[0, -X, 0]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	`[6,7,14,15]`
5	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[9,12,21,24]`
6	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	`[10,11,22,23]`

Table 36: Wyckoff bond: `48c@24g`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,17]`
2	$[-Y, -X, X]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[2,19]`
3	$[-Y, X, -X]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[3,18]`
4	$[Y, -X, -X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[4,20]`
5	$[X, Y, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,13]`
6	$[X, -Y, -X]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	`[6,15]`

continued ...

Table 36

No.	vector	center	mapping
7	$[-X, -Y, X]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[7,14]
8	$[-X, Y, -X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[8,16]
9	$[X, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9,21]
10	$[-X, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[10,23]
11	$[X, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[11,22]
12	$[-X, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[12,24]

Table 37: Wyckoff bond: 48d@24g, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[10,-11]
7	$[X, 0, Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[13,-16]
8	$[-X, 0, Y]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[14,-15]
9	$[0, Y, X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[17,-20]
10	$[0, Y, -X]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[22,-23]

Table 38: Wyckoff bond: 96e@24g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[12]
13	$[Y, X, Z]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[13]
14	$[-Y, -X, Z]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[14]
15	$[Y, -X, -Z]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[15]
16	$[-Y, X, -Z]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[16]
17	$[X, Z, Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[17]
18	$[-X, Z, -Y]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[18]

continued ...

Table 38

No.	vector	center	mapping
19	$[-X, -Z, Y]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[19]
20	$[X, -Z, -Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[20]
21	$[Z, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[21]
22	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[22]
23	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[23]
24	$[-Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[24]

* Wyckoff site: 48h, site symmetry: ..m

Table 39: Wyckoff bond: 48a@48h, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, z]$	[1,13]
2	$[-X, -X, Y]$	$[-x, -x, z]$	[2,14]
3	$[-X, X, -Y]$	$[-x, x, -z]$	[3,16]
4	$[X, -X, -Y]$	$[x, -x, -z]$	[4,15]
5	$[Y, X, X]$	$[z, x, x]$	[5,21]
6	$[Y, -X, -X]$	$[z, -x, -x]$	[6,22]
7	$[-Y, -X, X]$	$[-z, -x, x]$	[7,24]
8	$[-Y, X, -X]$	$[-z, x, -x]$	[8,23]
9	$[X, Y, X]$	$[x, z, x]$	[9,17]
10	$[-X, Y, -X]$	$[-x, z, -x]$	[10,18]
11	$[X, -Y, -X]$	$[x, -z, -x]$	[11,20]
12	$[-X, -Y, X]$	$[-x, -z, x]$	[12,19]

Table 40: Wyckoff bond: 48b@48h, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	[1,-13]
2	$[-X, X, 0]$	$[-x, -x, z]$	[2,-14]
3	$[-X, -X, 0]$	$[-x, x, -z]$	[3,-16]
4	$[X, X, 0]$	$[x, -x, -z]$	[4,-15]
5	$[0, X, -X]$	$[z, x, x]$	[5,-21]
6	$[0, -X, X]$	$[z, -x, -x]$	[6,-22]
7	$[0, -X, -X]$	$[-z, -x, x]$	[7,-24]
8	$[0, X, X]$	$[-z, x, -x]$	[8,-23]
9	$[-X, 0, X]$	$[x, z, x]$	[9,-17]
10	$[X, 0, -X]$	$[-x, z, -x]$	[10,-18]
11	$[-X, 0, -X]$	$[x, -z, -x]$	[11,-20]
12	$[X, 0, X]$	$[-x, -z, x]$	[12,-19]

Table 41: Wyckoff bond: `96c@48h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, z]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -x, -z]$	`[4]`
5	$[Z, X, Y]$	$[z, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$[z, -x, -x]$	`[6]`
7	$[-Z, -X, Y]$	$[-z, -x, x]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, x, -x]$	`[8]`
9	$[Y, Z, X]$	$[x, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-x, z, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[x, -z, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-x, -z, x]$	`[12]`
13	$[Y, X, Z]$	$[x, x, z]$	`[13]`
14	$[-Y, -X, Z]$	$[-x, -x, z]$	`[14]`
15	$[Y, -X, -Z]$	$[x, -x, -z]$	`[15]`
16	$[-Y, X, -Z]$	$[-x, x, -z]$	`[16]`
17	$[X, Z, Y]$	$[x, z, x]$	`[17]`
18	$[-X, Z, -Y]$	$[-x, z, -x]$	`[18]`
19	$[-X, -Z, Y]$	$[-x, -z, x]$	`[19]`
20	$[X, -Z, -Y]$	$[x, -z, -x]$	`[20]`
21	$[Z, Y, X]$	$[z, x, x]$	`[21]`
22	$[Z, -Y, -X]$	$[z, -x, -x]$	`[22]`
23	$[-Z, Y, -X]$	$[-z, x, -x]$	`[23]`
24	$[-Z, -Y, X]$	$[-z, -x, x]$	`[24]`

* Wyckoff site: `96i`, site symmetry: `1`

Table 42: Wyckoff bond: `96a@96i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, y, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -y, -z]$	`[4]`
5	$[Z, X, Y]$	$[z, x, y]$	`[5]`
6	$[Z, -X, -Y]$	$[z, -x, -y]$	`[6]`
7	$[-Z, -X, Y]$	$[-z, -x, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, x, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, z, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -z, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -z, x]$	`[12]`
13	$[Y, X, Z]$	$[y, x, z]$	`[13]`
14	$[-Y, -X, Z]$	$[-y, -x, z]$	`[14]`
15	$[Y, -X, -Z]$	$[y, -x, -z]$	`[15]`
16	$[-Y, X, -Z]$	$[-y, x, -z]$	`[16]`

continued ...

Table 42

No.	vector	center	mapping
17	$[X, Z, Y]$	$[x, z, y]$	[17]
18	$[-X, Z, -Y]$	$[-x, z, -y]$	[18]
19	$[-X, -Z, Y]$	$[-x, -z, y]$	[19]
20	$[X, -Z, -Y]$	$[x, -z, -y]$	[20]
21	$[Z, Y, X]$	$[z, y, x]$	[21]
22	$[Z, -Y, -X]$	$[z, -y, -x]$	[22]
23	$[-Z, Y, -X]$	$[-z, y, -x]$	[23]
24	$[-Z, -Y, X]$	$[-z, -y, x]$	[24]